

Penetration Testing Report

Target Information

- **URL**: http://dvwa:80
- **IP Address**: 172.18.0.3
- **Hostname**: dvwa
- **Service**: Apache httpd 2.4.25 ((Debian))

Executive Summary

The target host `dvwa` is running an outdated version of the Damn Vulnerable Web Application (DVWA) on

Findings

1. Service Enumeration (Nmap)

The target hosts an Apache web server running version 2.4.25 on port 80. This version is known to have s

- **Severity**: Low/Medium
- **Evidence**: Nmap service detection confirmed `Apache httpd 2.4.25`.

2. Web Vulnerabilities (Nikto)

A comprehensive scan using Nikto revealed multiple issues:

- **Directory Traversal**: The application allows access to sensitive files via path traversal attacks (e.g., ac
- **Unsafe Headers**: Multiple Content-Type header injection vulnerabilities were detected, allowing potent
- **Outdated Software**: The server software version is outdated and lacks security patches.
- **Severity**: High
- **Evidence**: Nikto reported 16 errors and items across 7799 requests, specifically flagging directory trav

3. SQL Injection (sqlmap)

The application is susceptible to SQL Injection attacks.

- **Severity**: Critical

- **Evidence**: sqlmap successfully identified a database management system and extracted data using S

Recommendations

1. **Patch or Upgrade**: Update Apache httpd to the latest stable version immediately.
2. **Input Validation**: Implement strict input validation and parameterized queries to mitigate SQL Injection.
3. **Access Control**: Restrict access to sensitive system files to prevent directory traversal.
4. **Header Security**: Sanitize HTTP headers to prevent injection attacks.
5. **Isolation**: If this is a production environment, isolate the host or decommission it as it poses an unacco